

Unit - 2

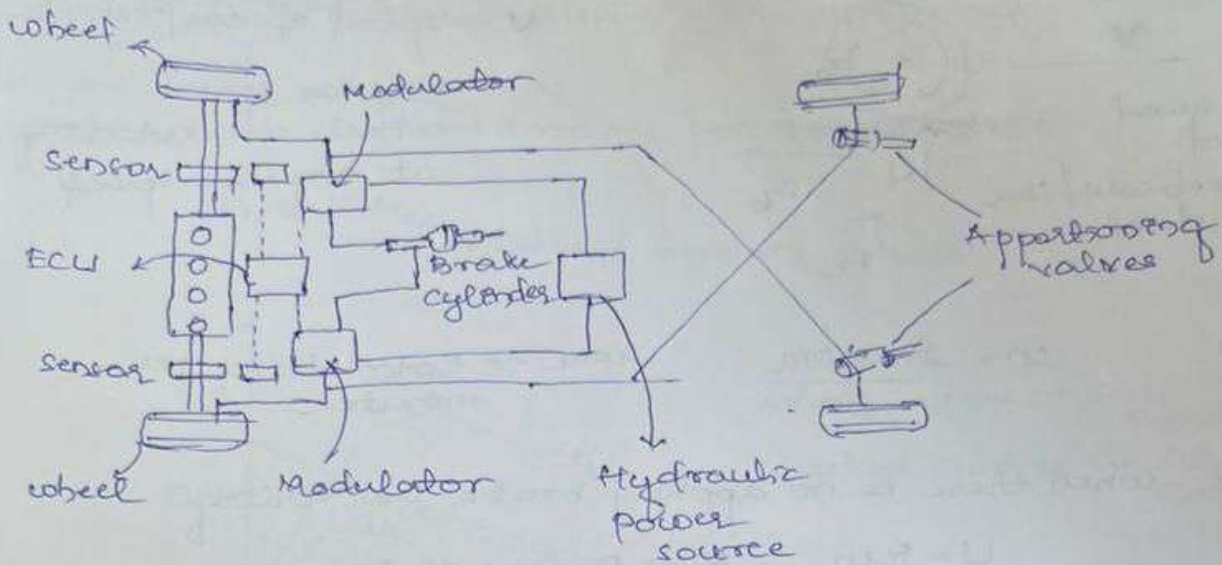
Chapter - 3 Vehicle safety and stability systems

Syllabus

- * Passive / Active safety
- * ABS and Problems on Slip
- * Traction Control
- * Electronic stability Program (ESP)
- * Electronic suspension system
- * Intelligent cruise control
- * Steering control systems
- * Airbag

Anti-lock Braking System

①



Components

- * Brake peddle
- * Master cylinder
- * Vacuum Boost
- * wheel cylinder
- * Caliper/disks
- * Brake lines

Scenario

- * ABS is a safety-related feature that assists the driver in deceleration of the vehicle in poor or marginal braking conditions (e.g. wet or icy roads)
- * Panic braking by the driver results in reduced braking effectiveness and typically, loss of directional control due to tendency of the wheels to lock.

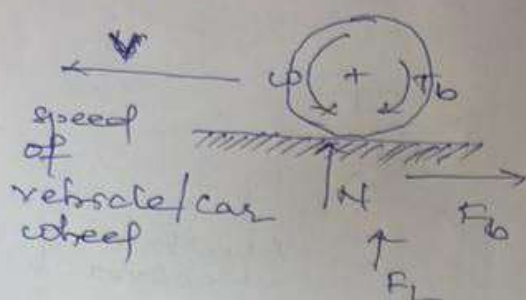
Why?

- * ABS allows the driver to maintain steering control of vehicle while in hard braking situations.

Motivation

- Under hard braking, an ideal braking system should
 - * provide the shortest stopping distances on all surfaces
 - * Maintain vehicle stability and steerability

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$V \rightarrow$ speed of car/vehicle
 $\omega \rightarrow$ Angular speed i.e. wheels are rotating at angular speed

$$\omega = \frac{2\pi \text{ RPM}}{60}$$

RPM $\rightarrow$ Rotations per minute

when there is no applied brake i.e. rolling
 $U = R\omega$ $R \rightarrow$ Radius of Tire

- * When brake pedal is depressed, force acts as Torque T_b in opposition to the wheel rotation.
- * The actual force that decelerates the car F_b
- * F_L is lateral force that maintains directional control of the car.
- * The angular speed of wheel begins to decrease, causing a difference between vehicle speed V and tire speed over road (i.e. ωR) $\rightarrow U$

$$U = \omega R$$

Slip as a percentage of car speed

$$S = \frac{V - U}{V} \times 100\%$$

A rolling tire has slip $S=0$ and a fully locked tire has $S=100\%$

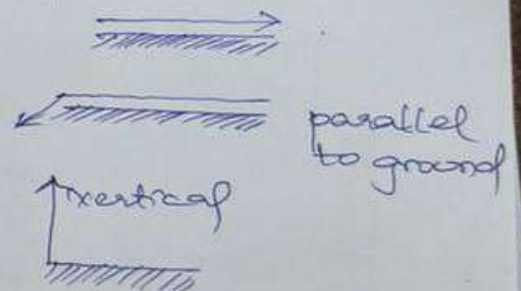
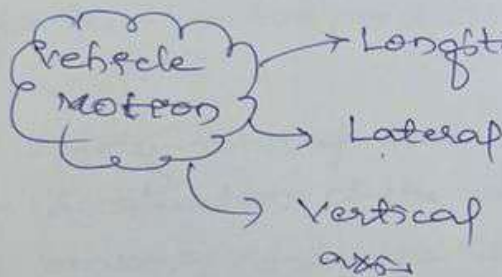
Passive safety Features

- ① Airbags
- ② seat belts pre-tensioners
- ③ Reinforced passenger cell
- ④ Laminated Glass



Active safety Features

- ① Anti-lock Braking system (ABS)
- ② steering control
- ③ suspension system
- ④ Autonomous emergency braking
- ⑤ Lane Departure
- ⑥ Traction control system (TCS)
- ⑦ TPMS (Tire pressure monitoring system)



- ① Translational
- ② Rotational

Displacement & Angular

Rotation of vehicle wheel

Angular displacement about longitudinal

Roll

Yaw
about vertical axis

Pitch

Lateral axis

- * For wet or icy roads, μ at $s = 100\%$ is so low that the lateral force is insufficient to maintain directional control of vehicles.
- * Directional control can often be maintained even in poor braking conditions if slip is optimally controlled.
- * In ABS equipped cars under marginal or poor braking conditions, the driver simply applies a steady brake force and system adjusts tire slip to optimum value automatically.

Why slip is optimal value?

① At $s = 0\%$ - wheels are rolling freely

② Why not high?

i.e. $s > 20-30$

- Tire enters skidding region
- Friction drops from static to kinetic friction that results in
 - * longer stopping distance
 - * Loss of steering control

③ $s = 10$ to 20% is optimal

At this range,

- * Tire operates near peak friction

Given

- Maximum braking force
- Shortest stopping distance
- Good steering stability

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Role of ABS

- * Monitors wheel speeds
- * Detects excessive slip
- * Reduces brake pressure when slip is too high
- * Reapplies pressure when wheel regains speed

ABC Actuation

Estimating torque T_{w0} applied to the wheel at the road surface by braking force

$$T_{w0} = R F_b \rightarrow \text{Braking force}$$

$$T_b = K_b p \rightarrow \text{Brake pressure}$$

↙ Brake constant

↘ opposition to T_{w0} is braking torque T_b applied to the disk by calipers in response to brake pressure p

Difference between two torques acts to decelerate the wheel. In accordance with basic Newtonian mechanics the wheel torque T_{w0} is related to braking torque and wheel deceleration

$$\text{i.e. } T_{w0} = T_b + I_{w0} \frac{d\omega}{dt}$$

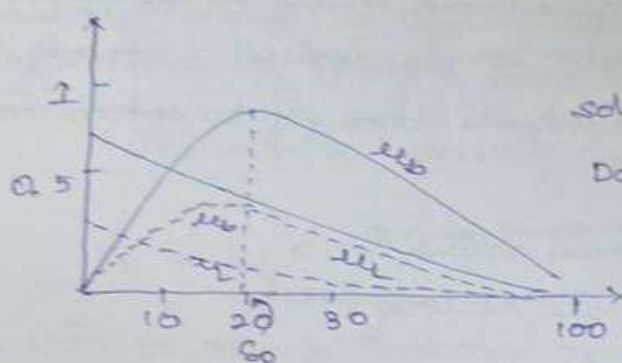
↙ wheel moment of inertia ↘ wheel deceleration

Brake pressure applied T_b w/ causing to slip increase, wheel torque is proportional to μ_b which reaches a peak at slip S_0 .

Braking force and Lateral forces are proportional to normal force acting on the tire/road interface

$$F_b = N \mu_b \rightarrow \text{braking friction co-efft}$$

$$F_L = N \mu_L \rightarrow \text{lateral friction co-efft}$$



Solid Curves $\rightarrow$ Dry Road
Dashed Curves $\rightarrow$ Icy Road

Graph $\rightarrow$ Braking co-efft v/c Tire slip

Case 1 $\rightarrow$ Dry Road / Dry Road without ABS
dry Road
As brake pedal force is increased from 0, slip also increases from 0

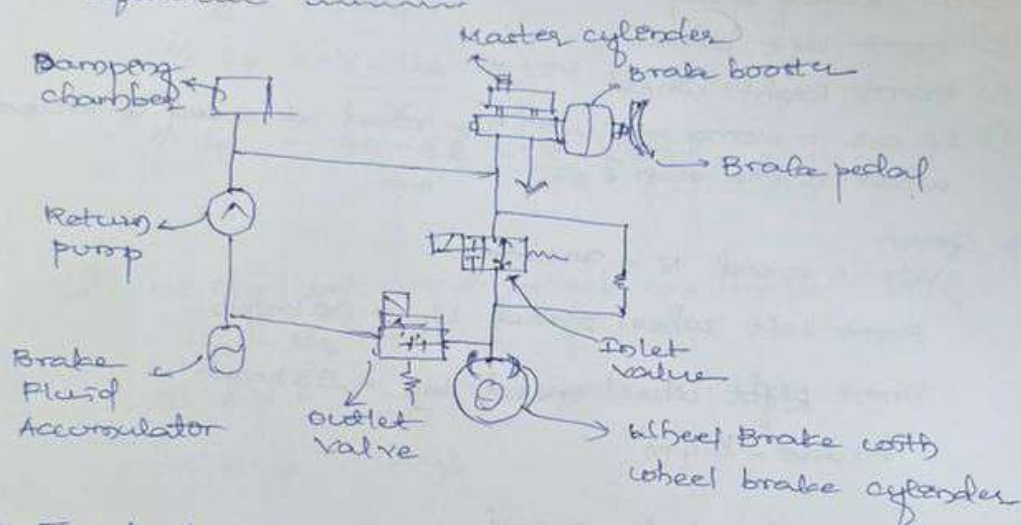
$s \propto \mu_b$ till $s = s_0$ $s_0 \rightarrow$ optimum slip

$s \propto \mu_L$ after $s = s_0$

μ_L decreases steadily with increases in the slip such that for fully locked wheels, the lateral force has its lowest value.

* It should be noted that by maintaining slip at s_0 , the maximum deceleration is achieved for a given set of conditions. Some reduction in lateral force occurs from its maximum by maintaining slip at s_0 . However, in most cases the lateral force is large enough to maintain directional control.

Hydraulic Modulator



- * The hydraulic modulator incorporates a series of solenoid valves that can open or close the hydraulic circuits between master cylinder and the brakes.
- * It can connect the brakes to the return pump, solenoid valves with two hydraulic connections and two valve positions are used.
- * The inlet valve between the master and brake controls applied pressure while the outlet valve between brake and return pump controls pressure release.
- * There is one pair of solenoid valves for each brake.

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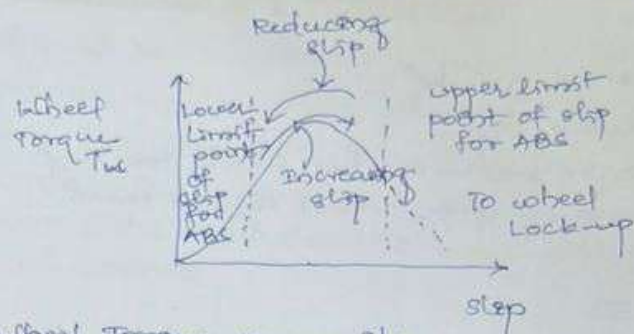
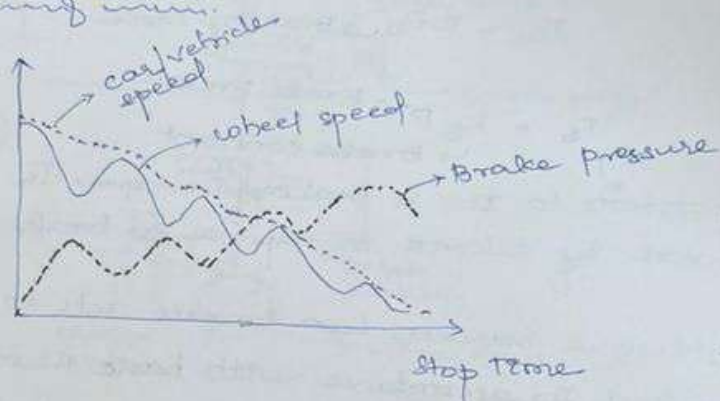


Fig Wheel Torque versus Slip

ABS Braking Action



- * Take an example that a car/vehicle is travelling at 55 kmph and the brakes are applied as indicated by rising brake pressure. The wheel speed begins to drop until the slip limit is reached.
- * At this point, ABS reduces brake pressure and wheel speed increases.
- * With the high applied brake pressure, wheels again tend toward lock-up and ABS reduces brake pressure the cycle continues until the vehicle is stopped.

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(b) Given $U_L = 68 \text{ kmph}$

$$\% S_L = \frac{V - U_L}{V} \times 100$$

$$\% S_L = \frac{70 - 68}{70} \times 100 = \frac{2}{70} \times 100 = \frac{200}{70} = 2.85\%$$

(c) $U_R = 63 \text{ kmph}$

$$\% S_R = \frac{V - U_R}{V} \times 100$$

$$\% S_R = \frac{70 - 63}{70} \times 100 = \frac{7}{70} \times 100 = 10\%$$

(d) on dry/wet roads, wheels are freely rotating

$$U_R = U_L = 0$$

$$\% \text{ slip} = \frac{V - U_R/U_L}{V} \times 100$$

$$\% \text{ slipp} = 100\%$$

Type 2

Calculate the wheel slip for F_L (Front Left) and Front right (F_R) for which the wheel rpm is measured as 450 rpm and 420 rpm respectively while radius is 0.4 m. When the vehicle speed is 70 kmph. Calculate slip

(a) At steady state speed

(b) Front wheel left

(c) Front right wheel

(d) If car is moving at the speed of 100 kmph on wet/dry road. what is the slip?

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Solⁿ → Given

$$\text{RPM of FL} = 450$$

$$\text{RPM of FR} = 420$$

$$\text{radius } R = 0.4$$

$$V = 70 \text{ kmph}$$

Here, First convert RPM into m/s or kmph

Front Left wheel, In this case, ^{type of problem} convert into kmph because vehicle speed is given in kmph

$$U_{\text{kmph FL}} = \frac{\omega R \times 3600}{1000}$$
$$= \frac{2\pi \text{RPM} \times R \times 3600}{60 \times 1000}$$

$$U_{\text{FL}} = \frac{2\pi \times 450 \times 0.4 \times 3600}{60 \times 1000}$$

$$U_{\text{FL}} = 67.85 \text{ kmph}$$

$$U_{\text{FR}} = \frac{\omega R \times 3600}{1000}$$

$$= \frac{2\pi \text{RPM} \times R \times 3600}{60 \times 1000}$$

$$U_{\text{FR}} = \frac{2\pi \times 420 \times 0.4 \times 3600}{60 \times 1000}$$

$$U_{\text{FR}} = 63.334 \text{ kmph}$$

Problems on slip

Type 1

- ① Calculate the wheel slip for FL (Front Left) and FR (Front Right), for which the wheel speed 68 kmph and 63 kmph respectively while the radius is 0.4 m. When the vehicle speed is 70 kmph. Calculate slip.
- ② At steady state speed
- ③ Front left wheel
- ④ Front Right wheel
- ⑤ If car moving on wet/icy road at speed of 100 kmph what is the slip?

Sol 2 → Given

Vehicle speed $V = 70 \text{ kmph}$

Front Left wheel speed $W_L = 68 \text{ kmph}$

Front Right wheel speed $W_R = 63 \text{ kmph}$

radius = 0.4 m

- ② At steady state speed
- $$V = W_R = W_L = 70 \text{ kmph}$$

$$\% \text{ slip} = \frac{V - W}{V} \times 100\%$$

③

$$\% \text{ slip for Right wheel} = \frac{V - W_R}{V} = \frac{70 - 70}{70} = 0\%$$

$$\% \text{ slip for Left wheel} = \frac{V - W_L}{V} = \frac{70 - 70}{70} = 0\%$$

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Type-3 problem

During a controlled emergency brake testing on a dry test track, a vehicle equipped with ABS (Anti-lock Braking system) is travelling at 25 m/s . The ABS controller monitors the wheel speed sensors and notices the following

- * Front left wheel completes 28 full rotations in 2 seconds
- * Front right wheel completes circumferential distance of 40.2 metres in same 2 seconds.

The effective tire radius for both wheels is 0.36 m .

Assume no significant wheel lift or suspension dynamics

Find

- without directly calculating rpm, determine the linear (tire surface) speeds of each front wheel from the given data.
- Use the vehicle speed and derived tire surface speeds to compute the wheel slip percentage for both wheels.

Solⁿ →

Given

$$V = 25 \text{ m/s}$$

$$R = 0.36 \text{ m}$$

$$t = 2 \text{ s}$$

Note: → (i) In this problem no need of conversion as vehicle speed is in m/s

- In question they mentioned don't calculate RPM

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continue

⑨ Linear (tire surface) speeds

Tire circumference

$$C = 2\pi R$$

$$C = 2\pi (0.36) = 2.262m$$

① Front-left wheel

Rotations = 28 in 2 sec

distance covered $D_L = \text{Rotations} \times \text{circumference of Tire}$

$$D_L = 28 \times 2.262$$

$$D_L = 63.34m$$

calculate speed of Left wheel

$$U_L = \frac{D_L}{t} = \frac{63.34}{2} = 31.67 \text{ m/s}$$

② Front-right wheel

$D_R = 40.2m$ in 2s

speed of Right wheel

$$U_R = \frac{D_R}{t} = \frac{40.2}{2} = 20.1 \text{ m/s}$$

⑩ wheel slip percentage

① front-right wheel slip

$$\% \text{ slip}_R = \frac{V - U_R}{V} \times 100$$

$$= \frac{25 - 20.1}{25} \times 100 = 19.6\%$$

Typical value during ABS braking
Indicates controlled slip for maximum braking force.

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① At steady state speed

$$V = U_{PR} = U_{FL} = 70 \text{ kmph}$$

$$\% S = \frac{V - U_{PR} / U_{FL} \times 100}{V} = \frac{70 - 70 \times 100}{70} = 0\%$$

② $\% S_L = \frac{V - U_{FL}}{V} \times 100$

$$\% S_L = \frac{70 - 67.85}{70} \times 100$$

$$\% S_L = 3.07\%$$

③ $\% S_R = \frac{V - U_{PR}}{V} \times 100$

$$\% S_R = \frac{70 - 63.334}{70} \times 100$$

$$\% S_R = 9.52\%$$

④ on icy Road / wet road, wheels are freely rolling

$$U_R = U_L = 0$$

$$\% \text{slip} = \frac{V - U_R / U_L \times 100}{V}$$

$$\% \text{slip} = 100\%$$

Traction Control Systems

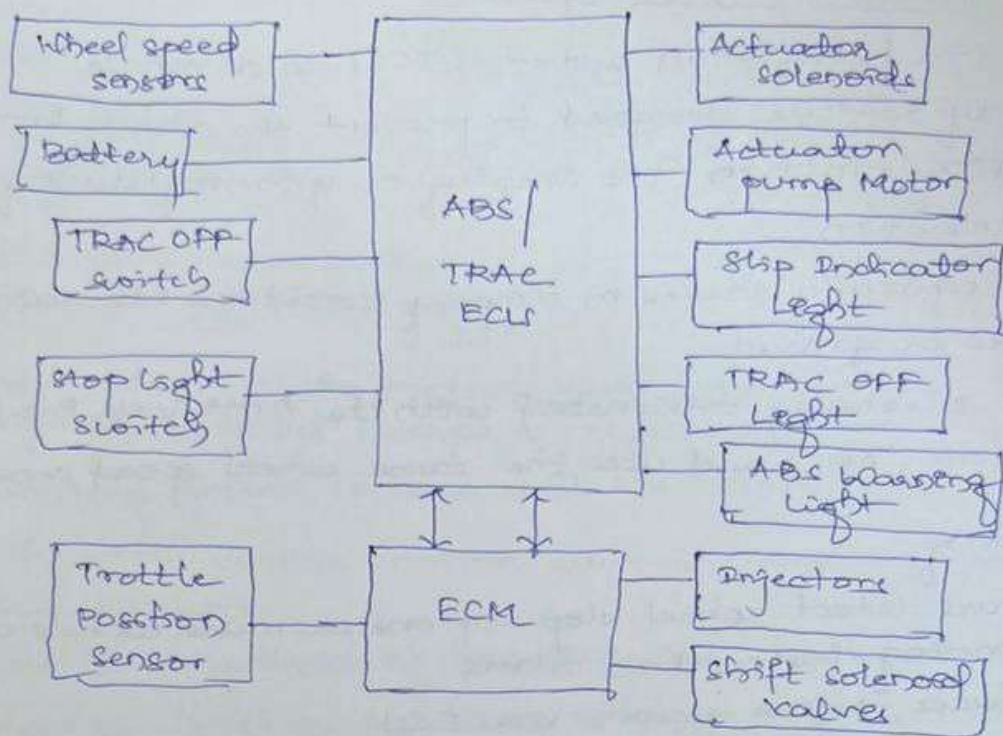
- * A traction control system (TCS) is a vehicle safety feature designed to prevent the wheels from losing traction (i.e. slipping or spinning) during acceleration
- * It's especially useful in slippery conditions like rain, snow or gravel.
- * TCS is usually integrated with the Anti-lock Braking system (ABS) and uses the same wheel speed sensors.

Working

- * Sensors detect wheel slip - if one or more wheels are spinning faster than others
- * Reduce engine power via ECU
- * Apply brake force to the slipping wheels

Traction Control system consists

- ① sensors
 - Left-Front
 - Left-Rear
 - Right-Front
 - Right-Rear
- ② Master cylinders
- ③ Amber ABS warning light
- ④ ABS Relays
- ⑤ Electronic brake control module (EBCM)
- ⑥ Pressure modulator Valve (PMV) Assembly



⑪ Front-Left wheel

$$\% \text{ slip} = \frac{V - W_L}{V} \times 100$$

$$\% \text{ slip} = \frac{25 - 31.67}{25} \times 100$$

$$\% \text{ slip} = \frac{-6.67}{25} \times 100 = -26.7\%$$

Negative slip $\rightarrow$ wheels is rotating faster than vehicle speed
indicating that possible measurement inconsistency
or momentary traction gain

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Two classes of Electronic Suspension Systems

- ① Active → Power is added to suspension system
- ② Semi-Active via a hydraulic or pneumatic power source

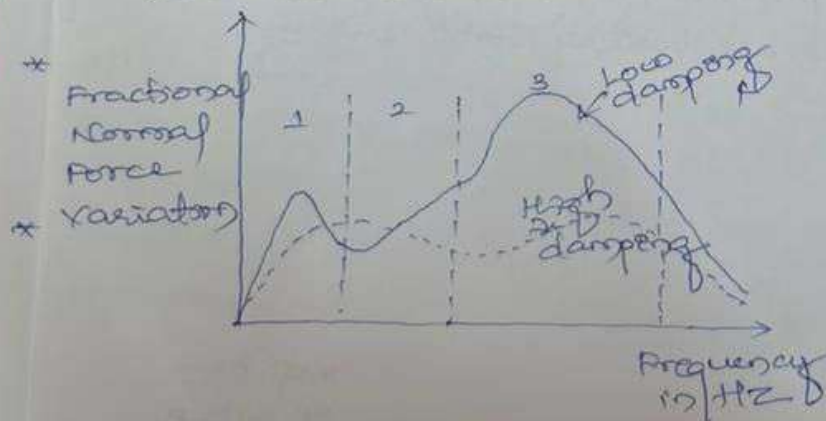
↓
purely dissipative (i.e. power is absorbed by shock absorber under control of microcontroller)

- * The shock absorber damping is regulated to absorb the power of wheel motion in accordance with driving conditions.

Semi-active suspension system (Regulates only damping)

- * A good ride for as much of the time as possible without sacrificing handling.
- * A Good ride is achieved if the car's body is isolated as possible from the road.
- * A semi-active suspension controls the shock absorber damping to achieve the best possible ride.
- * Suspension system has another major function. It must also dynamically maintain the tire normal force as sprung mass (wheel assembly) travels up and down due to road roughness.

Tire Force Variation



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Region	Frequency	Damping
1) Sprung mass mode	1-2	High
2) Intermediate axle	2-8	Low → Good subjective ride
3) Unsprung mass resonance	8-20	High
4) Harshness	>20	Low

- * For lower damping, the unsprung mass moves relatively freely due to road input while the sprung mass motion is relatively low.
- * Low damping results in relatively high variation in normal force particularly near to two peak frequencies, lower damping results in relatively poor handling characteristics.
- * Another major input to the vehicle that affects handling is steering input that causes maneuvers parallel to the road surface.
- * Whenever the car is executing such maneuvers, there is a lateral acceleration.
- * This acceleration acting through the center of gravity causes the vehicle to roll in a direction opposite to the maneuvers.
- * Car handling generally improves if the amount of roll for any given maneuver is reduced.
- * Rolling rate for a given car and maneuver is improved if spring rate and shock absorber damping are increased.
- * Handling is improved by increasing shock absorber damping as lateral acceleration increases.

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Electronic Suspension System

①

Purpose

- * To isolate the car body motion as much as possible from wheel motion due to rough road input.

Performance Measure

① Ride ② Handling

- * Ride refers to the motion of the car body in response to road bumps or irregularities.
- * Handling refers to how well the car body responds to dynamic vehicle motion such as cornering and hard braking.
- * Ride is improved by lowering the shock absorber damping whereas handling is improved by increasing the shock absorber damping.
- * In traditional suspension design, the damping parameter is fixed and is chosen to achieve a compromise between ride and handling.
- * In electronically controlled suspension systems, absorber damping can be varied depending on driving conditions and road roughness characteristics i.e. the suspension system adapts to inputs to maintain the best possible ride subject to handling constraints that are associated with safety.

Variable Damping via Variable strut fluid viscosity

- * Damping of a suspension system is determined by the viscosity of fluid in the shock absorber/strut and by the size of the aperture through which the fluid flows as the wheel moves relative to the car body.
- * For achieving variable damping is to switch between two aperture sizes using a solenoid.

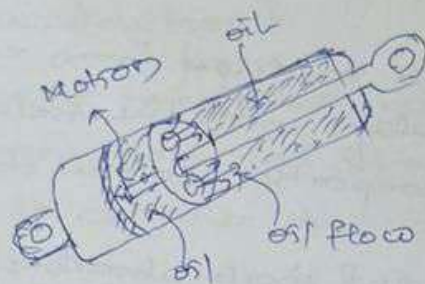


Fig → Adjustable shock Absorber

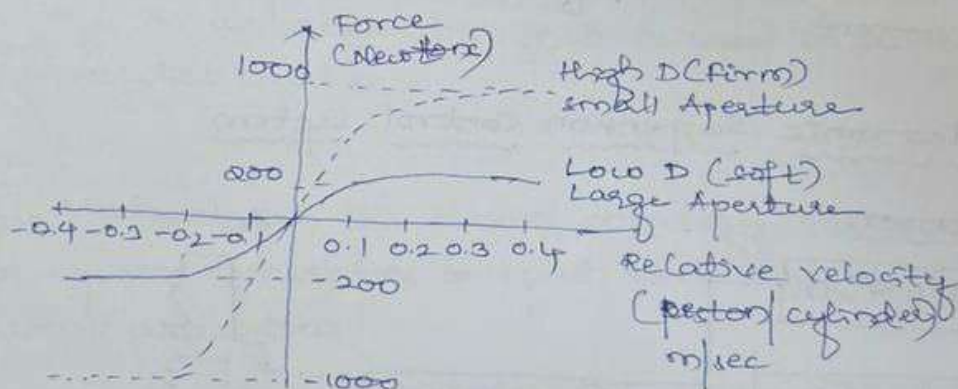


Fig → Force versus Relative velocity of a solenoid-switched Aperture shock Absorber

- * Variable suspension damping is achieved with a fixed aperture and variable fluid viscosity.
- * The fluid for such a system consists of a synthetic hydrocarbon with suspended iron particles.

- * An electromagnet is positioned such that a magnetic field is created whose strength is proportional to current through the coil.
- * This magnetic field passes through the magneto-rheological fluid (MR).
- * In the absence of magnetic field, the iron particles are randomly distributed and the MR fluid has relatively low viscosity corresponding to low damping.
- * As the magnetic field is increased from zero, the iron particles begin to align with the field and viscosity increases in proportion to the strength of field.
- * Damping of the associated shock absorber/strut varies continuously with the electromagnet coil current.

Electronic Suspension Control System

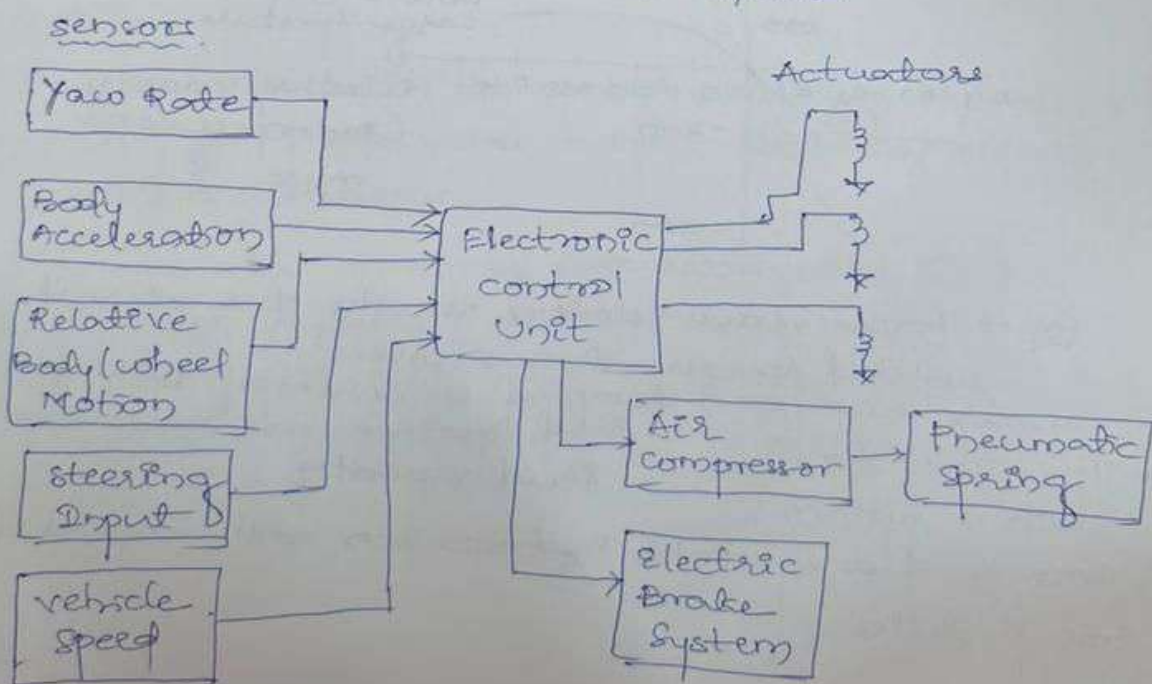


Fig. Control system for a typical Electronic Suspension system

Lateral acceleration $A_L \propto$ (proportional) to vehicle speed and input steering angle ④

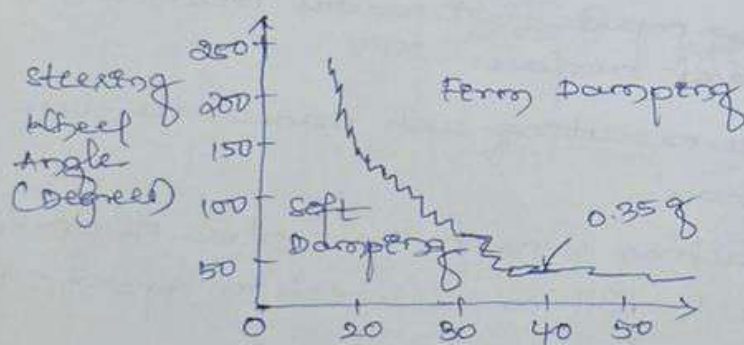
$$A_L = k V \delta$$

vehicle speed steering angle δ

The dynamics of a spring/mass/damping system, identifying resonant frequency and critical damping D_c

$$D_c = d \sqrt{k/m}$$

- * For good ride, the damping should be as low as possible.
- * For practical design considerations, the minimum damping is generally in the region of $0.1 < D/D_c < 0.2$
- * For optimum handling, the damping is in the region of $0.6 < D/D_c < 0.8$.



Practical considerations

- * In above explanation all the inputs are taken separately.
- * When driving along a nominally straight road with small steering inputs, the road is dominant.
- * In above case, the control is based on the spectral content i.e. frequency region of relative motion. The controller calculates the spectrum of relative velocity of the sprung and unsprung mass at each wheel.
- * The weighted amplitude of the spectrum near the peak frequencies exceeds a threshold, damping is increased, yielding a firmer ride and improved handling. Otherwise, damping is kept low (soft suspension).
- * Another possible strategy is possible via with motor-driven variable-aperture shock absorbers. Here, the shock absorber apertures are adjusted to minimize sprung mass acceleration in the 2 to 8 Hz frequency region for providing optimum ride control.
- * Large steering input $\rightarrow$ switches to small aperture yielding stiffer suspension and improved handling.

* The control system typically is in the form of a microcontroller or microprocessor based digital controller.

* The outputs from each sensor are sampled, converted to digital format and stored in memory.

Sensors

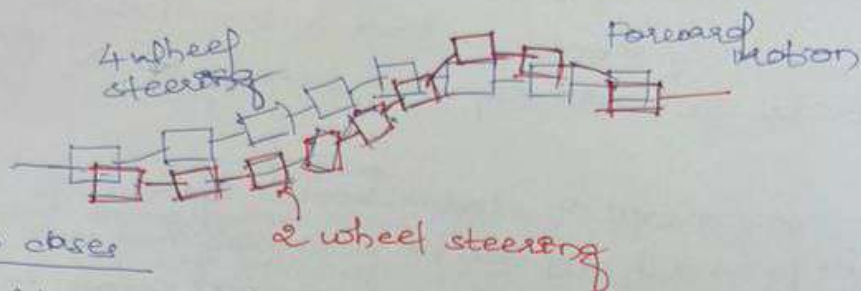
- ① The body acceleration measurement can be used to evaluate ride quality. The controller does this by computing a weighted average of the spectrum of the acceleration.
- ② The relative body/wheel motion can be used to estimate the normal force and damping is then adjusted to try to optimize this normal force.
- ③ The yaw rate sensor provides data which in relation to vehicle speed and steering input measurement is combined in an algorithm that is used to activate the ^{electro}hydraulic brakes.

Actuators

- * In accordance with the control strategy, the electronic control system generates output electrical signals to various actuators.
- * The variable damping actuators vary either the oil passage orifice or the RH fluid viscosity independently at each wheel to obtain the desired damping for that wheel.

Lane change maneuvers

(2)



- ① At low speeds $\rightarrow$ Maneuvering ability is enhanced
- * The rear steering angle is in the opposite direction to the front steering angle.
 - * The control strategy has the effect of decreasing the car's turning radius by as much as 30% from the value it has for front wheel steering only.

- ② At intermediate speeds, the steering might be front wheel only

- ③ At higher speeds
- * Front wheels and rear wheels are steered in same direction.
 - * At highway cruise speed, the rear wheels turn in the opposite direction to the front wheels for a very short period and then turn in the same direction as the front wheels.

* When front wheels and rear wheels turn in same direction means

- Angle between car & trailer axes is less than it is for front wheel steering only.
- Reduction in angle means that the lateral force applied to the rear wheels by the trailer in curves is less than that for front wheel
- Lateral force reduction improves stability of the vehicle

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Airbag

Supplementary Restraint System for driver and/or passenger safety in case of a crash.

Why do we need airbags?

- * Federal Government mandate in 1989 states that we need some sort of automatic occupant safety device.

choice 1 is an automatic safety belt feature

- * very costly
- * high Failure Rate

choice 2 Air-Bag

- * Universal Production
- * Better safety choice

choice 3 Both (Best)

Components of Airbag system

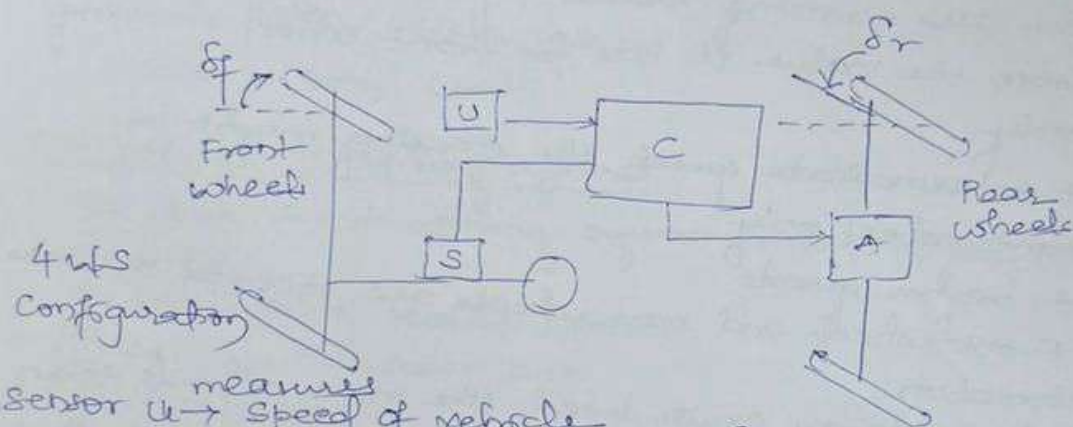
The dual front airbag system consists of the following components:

- (i) Air control Module (ACM)
- (ii) Airbag indicator lamp
- (iii) clockspring
- (iv) Driver and passenger side airbag modules
- (v) Driver and passenger side knee blockers
- (vi) wire harness and connections.

clockspring → Located inside the ^{steering} wheel. This device is used to connect the hard wired airbag harness to the system harness while allowing the steering wheel to maintain a full range of motion.

Electronic Steering Control

- * Power steering scheme utilizes a special electric motor to provide the boost required instead of the hydraulic boost.
- * Electronic control of electric boost systems is straightforward and can be accomplished without any energy conversion from electrical power to mechanical actuation.
- * Electronic control offers very sophisticated adaptive control in which the system can adapt to the driving environment.



4WS Configuration

Sensor $U \rightarrow$ Speed of vehicle
 Sensor $S \rightarrow$ Steering angle
 ecc $C \rightarrow$ Microcontroller

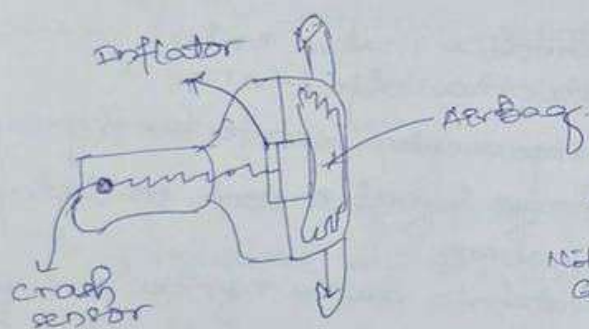
$\delta_f \rightarrow$ front wheel steering angle
 $\delta_r \rightarrow$ rear steering angle

- * The front wheels are steered to a steering angle δ_f by the driver's steering wheel input. The controller determines the desired rear steering angle δ_r under program control as a function of speed and front steering angle.

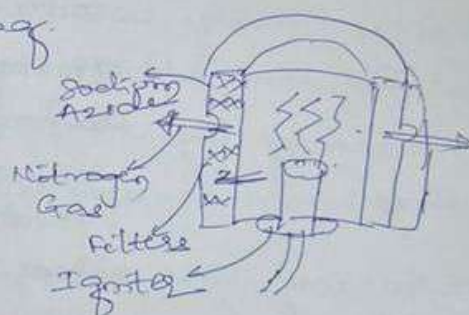
Working Principle of Airbag

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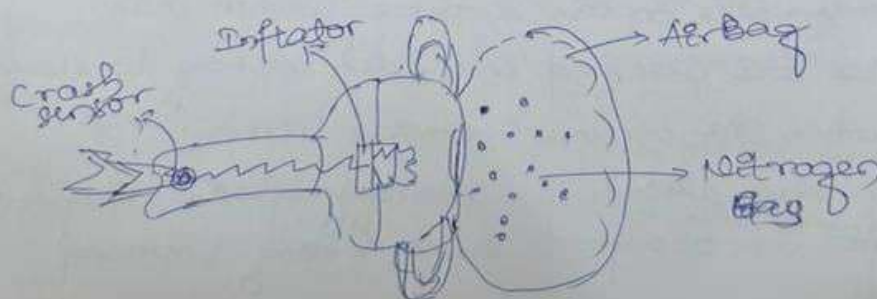
- * A thin nylon bag in the steering wheel inflates in the event of an impact and prevents the driver/passenger from hitting the steering wheel/dashboard.
- * Several crash sensors are located in the front of vehicle and in the passenger compartment.
- * Each senses the sudden deceleration or impact in the event of a crash and flips a mechanical switch to indicate a crash.
- * The sensors can detect the direction of impact, so front airbags will generally not deploy during side impacts. In modern vehicles side airbags are deployed.



Front Airbag in steering wheel



Airbag Inflation device



Front Airbag After collision

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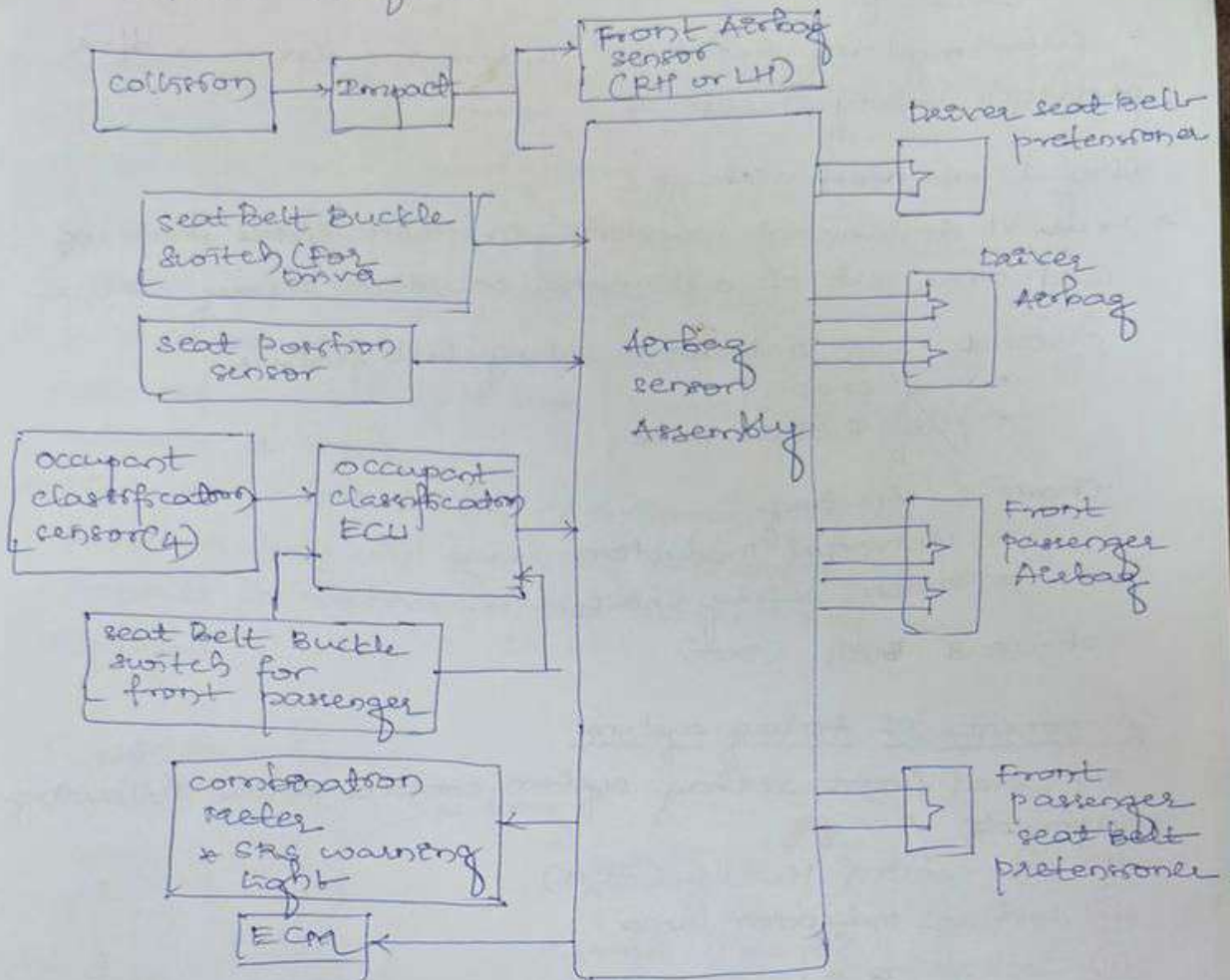
- * After collision, sensor sends an electric current to igniter system.
- * The electric current is used to heat a filament wire that ignites the NaN_3 (Sodium Azide) capsules producing N_2 .
- * 130gm of NaN_3 produces 60 litres of N_2 gas.
- * The gas expands quickly and inflation of airbag takes place, which breaks through a plastic cover in the steering wheel.
- * Airbags are designed to inflate in frontal impacts in which the car strikes about 16km/hr.

Time in milliseconds	Actions
0	Automobile makes initial contact with other object
30	Microprocessor decides to deploy airbag. Signal is sent to deployment mechanism
32	Pyrotechnic device ignites and gas enters airbag
60	Airbag is fully inflated.

- * Immediately after full inflation, the airbag deflates through tiny pores on the surface within 0.3s.
- * Several holes are designed into each airbag to slowly expel gas when the occupant strikes airbag.
- * This action helps absorb the energy of the occupant and prevent the occupant from being bounced backwards.

System Diagram

②



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Seat position sensor

- * The seat position sensor which is attached to the seat rail of the driver seat detects the sliding position of the seat. In addition, the shielding plate is installed on the seat rail to make the seat position sensor judge the seat position.
- * The seat position sensor uses a Hall IC for its sensor and has a magnet position on its opposite side.
- * The seat position detection by seat position sensor judges that
 - (i) seat position is rearward if the shielding plate is between the sensor.
 - (ii) The seat position is forward if the shielding plate is not between the sensor.

Sensors used by the Airbag unit

- ① Occupant Detector
- ② Seat Belt Pretensioner
- ③ Crash sensor
- ④ Seat position sensor
- ⑤ Safety sensor

Seat Belt Pretensioner Operation

- * When the pretensioner is inactivated, the clearance between the roller and the sleeve is secured and thus the sleeve rotates freely. The spool ^{with} which winds the seatbelt rotates together ~~as~~ ^{with} the sleeve.
- * The frontal collision, SRS-ECU which has detected the impact exceeding the threshold from the front impact sensor is energized to ignite gas generator. Then gas is generated, and the gas pressure moves up the piston.
- * When the piston moves up, the rack gear of the piston is engaged with the pinion, gear of the pinion to rotate the pinion. When the pinion rotates, the pinion is tilted to move the roller to the center and the pinion, sleeve and spool are incorporated.
- * The gas pressure moves up the piston further and the pinion rotates the spool to wind the belt.